



United International University (UIU)

Dept. of Computer Science & Engineering (CSE)

Mid Term Exam Trimester: Spring 2026 Marks: 30 Time: 1 hr 30 mins

Code: CSE 3411 Course Title: System Analysis & Design

“Any examinee found adopting unfair means will be expelled from the trimester/program as per UIU disciplinary rules.”

Answer the following questions:

A university introduces an automated system to manage scholarship applications efficiently. Students log into the system and submit applications by providing academic records, financial documents, and a personal statement. The system performs basic validation (e.g., checking missing documents) and stores the data in the Scholarship Application Table, while student details are retrieved from the Student Profile Table.

After submission, the system performs an AI-based initial screening, assigning a score based on criteria such as academic performance and financial need. These scores are stored in the AI Evaluation Table. Applications that meet the minimum criteria are forwarded to faculty reviewers.

Faculty reviewers evaluate the applications and can approve, reject, or request additional documents. If additional documents are requested, the student is notified and can resubmit the application. All reviewer decisions and comments are recorded in the Faculty Review Table, and the application status is updated accordingly.

Approved applications are then sent to the Finance Office for verification of financial information. The results are stored in the Financial Verification Table. Applications that pass this stage are forwarded to the Scholarship Committee.

The Scholarship Committee reviews academic history from the Academic Record Table and past scholarship data from the Scholarship History Table before making a final decision. Final outcomes are recorded in the Final Decision Table.

If approved, the scholarship is granted and recorded in the Scholarship Award Table, and the student is notified. If rejected at any stage, the decision is recorded and the student is informed. Students may submit an appeal, which is stored in the Appeal Request Table and reviewed by the Appeals Board. Their decisions are recorded in the Appeal Decision Table, and notifications are sent to the student.

- a. Draw the Data Flow Diagram (DFD) for the above scenario. [3]
- b. Draw the Activity Diagram for reviewing a scholarship application. [3]
- c. Draw the Use Case Diagram (include at least one «include» or «extend» relationship). [3]
- d. Write the descriptive form of one major use case from your diagram. [3]

2. a. Considering the above-described scenario was in its initial stage before launch. Briefly explain any **one tool** of information gathering that would be **most beneficial** in gathering the information needed to make this project. Mention the **merits** and **demerits** of the tool you chose.

[1+2]

- b. Two students, Sara and Ahmed, are debating which requirements prioritization approach categorizes features based on customer satisfaction. Ahmed argues it is the MoSCoW method, while Sara believes it is the Kano Model - who is correct? Identify and briefly describe the model that categorizes features based on customer satisfaction.

[2] [CO2]

3. Answer the following questions:

[CO1]

- a. What is system boundary?

[1]

- b. A group of university students plans to build a campus lifestyle mobile app that includes features like class schedules, event updates, food delivery options, and a meme-sharing section. At the start, they are not sure about all the features because trends keep changing (for example, today students want a meme page, tomorrow they may want a study group feature). The developers frequently release small updates, share early versions of the app with classmates, and collect feedback through polls and comments. Based on this feedback, they keep modifying the app, adding new features, removing unpopular ones, and improving the user interface.

Which model would be more suitable for this scenario, Agile Model or Iterative Model? **Justify** your choice and **explain** why it is better than the other option.

[3]

- c. What is the full form of **MIS** and **DSS**?

[1]

4. A venture capital firm is assessing a luxury wellness retreat chain for potential funding. The project requires an initial investment of \$20,000 at present, followed by an additional investment of \$X at the end of Year 1, and a further expenditure of \$5,000 at the end of Year 2. The venture is expected to generate revenues of \$4,000 at the end of Year 2, \$9,000 at the end of Year 3, \$13,500 at the end of Year 4, and \$12,000 at the end of Year 5. Given that the net cash flow of this project is \$4,500, it determines the Year 1 investment value.

[3]

[CO4]

5. Develop and present in the classroom a **partial** SRS document (software requirement specification) as a group wise project assigned in the classroom. The SRS document should cover the following areas regarding your respective project: Introduction, Rationale/purposes of the project, Detail system study (tentatively you should cover benchmark products, research papers, onsite visit, survey, online sources etc.), feasibility analysis (SWOT analysis, cash flow analysis etc.), system design (Use case diagram, use case descriptive form, Data Flow Diagram, activity diagram, swim lane diagram etc.)

(do not answer this question in the examination)

[CO3]

[5]